

ZTIA Ecosystem Map: Zero Trust Identity Access, Layer by Layer

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The Core Thesis

Modern enterprise security isn't one product category — it's a stack of distinct layers, each solving a specific pain, each owned by different companies (sometimes the same company operating in more than one layer at once).

Zero Trust Identity Access (ZTIA) is the umbrella framework for organizing this stack. It answers three questions for any company, product, or capability:

1. **What pain does this actually solve?**
2. **What layer of the stack does it operate at?**
3. **What can it not do — and who fills that gap?**

This isn't a product catalog. A catalog lists what companies sell. An ecosystem map diagnoses what pain exists and who's actually solving each piece of it — including the reality that one company can show up in more than one layer, and that some layers currently have real, unfilled gaps.

Why Layers, Not Companies

Organizing by company category breaks down fast — most of the interesting companies right now don't stay in one lane:

- **Origin** looks like an Endpoint tool but its real differentiation is causal observability, not device management
- **Dosu** looks like a dev tool but structurally solves the same "discovery + context" problem as **Cyera** — just for knowledge instead of data
- **1Password** spans Secrets/Credentials AND, as of July 2026, Financial/Spend visibility — two different jobs, one company
- **LangChain** looks like it should govern agents but explicitly does not — it builds them

Organizing by **layer and pain** instead of company name is what makes this an actual ecosystem map rather than a vendor list.

The Full Layer Map

Layer	Core Question It Answers	Companies	Notes
Network	What can reach what?	Tailscale , Twingate, ZeroTier	Network-as-gate — identity-verified connectivity, segmentation, least-privilege reachability
Identity & Access	Should this identity (human or agent) be here?	Okta , Microsoft Entra ID, C1 (formerly ConductorOne) , Lumos	Access governance, SSO, lifecycle management, entitlement reviews. C1 has expanded well beyond this single layer since July 2026 — see deep dive
Discovery & Context — Data	Do we know what sensitive data we have, and is it exposed?	Cyera	DSPM — discovery, classification, risk scoring for data
Discovery & Context — Knowledge	Does this agent actually understand what it's looking at?	Dosu	Grounds AI answers in real project context to prevent hallucination
Detection	Is something behaving	Artemis , CrowdStrike, Splunk, SentinelOne	Cross-domain behavioral detection and response — this is the job, regardless of deployment mechanism.

Layer	Core Question It Answers	Companies	Notes
	wrong right now?		Splunk is the dominant legacy SIEM every AI-native player positions against; SentinelOne is one of three EDR vendors now acquiring pipeline tooling directly (see Data Pipeline & Routing)
Telemetry Visualization & Aggregation	Can we see and correlate signals across metrics, logs, and traces in one place?	Grafana Labs (LGTM stack)	General infrastructure observability, not identity-native — the layer other layers' data often gets visualized through; edging toward Detection as it adds SOC/security analytics capability
Data Pipeline & Routing	How does telemetry actually get from every source to every destination, in a usable shape?	Cribl	The pipe, not the destination — sits upstream of Telemetry Visualization, Detection, and every other layer that consumes telemetry. Vendor-neutral by design, though EDR vendors are increasingly acquiring competing pipeline tools to reduce dependency on neutral players
Agent Construction	How do we build the	LangChain	Orchestration/development layer — the raw material other layers react to.

Layer	Core Question It Answers	Companies	Notes
	agent in the first place?		Explicitly does NOT govern what it builds
Endpoint Visibility	What's on this machine, is it compliant, what changed?	Tanium, Ivanti, Axonius	State snapshots, inventory, query-based — tells you <i>that</i> something happened, not <i>why</i> . Axonius's actual scope extends beyond pure endpoint into full asset attack-surface visibility (CAASM) — devices, SaaS, cloud, identity, and OT/IoT
Hybrid Workforce Observability	Why did this happen — what did a human or AI agent actually do, step by step?	Origin	Vendor-agnostic causal tracing (prompt → reasoning → action → outcome) across both human and AI activity — the layer nobody else operates at with this depth
Secrets & Credentials	Are secrets and non-human credentials protected?	1Password , HashiCorp Vault, C1 (Agentic Vault)	Vaulting, rotation, NHI governance. C1's Agentic Vault (July 2026) is a genuine new entrant using workload federation instead of credential delivery
AI Financial Visibility	What are we actually spending on	1Password (AI Spend and Consumption Management)	Financial/spend visibility ONLY — explicitly not behavioral or causal. "Token bills tell you what you

Layer	Core Question It Answers	Companies	Notes
	AI, and by whom?		spent. [Hybrid Workforce Observability] tells you what you bought."
Cloud Posture	Is our cloud environment misconfigured or exposed?	Wiz, Orca	CSPM — scans cloud resource configs for exposure
IT Workflow Orchestration	How do we turn ground-truth data into automated action?	ServiceNow (Now Assist AI Agents; also ITOM AI Prime powered by Tanium)	Consumes endpoint/data truth from other layers, orchestrates workflows and remediation on top of it — does not generate the ground truth itself

Layer Deep Dives

Network

Network-as-gate, not network-as-sensor. Encryption has made deep packet inspection largely obsolete, but controlling *what can reach what* is more important than ever in a Zero Trust world.

Tailscale's specific differentiator: identity-first mesh networking at the IP layer, direct WireGuard tunnels, no chokepoints, no broadcast noise (unlike ZeroTier's Layer 2 emulated-switch model). ACLs enforce least-privilege reachability for both human and agentic connections (via tsnet).

What this layer cannot do: see what happens after a connection is made, govern SaaS OAuth tokens issued directly between browser and app, or inspect endpoint-level behavior.

Aperture vs. C1's identity-layer approach: Tailscale's Aperture gateway and C1's Agentic Vault/Runtime Governance both replace raw API keys with identity for AI/LLM traffic — same philosophy, opposite ends of the stack (network vs. identity platform). Not real substitutes; more a sign the industry is converging on "identity replaces secrets" independently. Full comparison in [tailscale-research.md](#) and [c1-research.md](#).

Identity & Access

The traditional core of IAM — but the agentic era breaks assumptions this layer was built on. Traditional IAM assumed identity = human, sessions are bounded, and accountability is inherent. Agents break all three: they're non-human identities, may run indefinitely, and require explicit audit design to be accountable at all.

Okta's agentic answer: three questions — where are my agents, what can they connect to, what can they do — anchored by discovery, least-privilege access, kill switches, and lifecycle governance.

C1 vs. Lumos — the closest head-to-head in this layer, both AI-native and fast-deploying against slow legacy IGA. Neither has a clean structural advantage; execution and integration fit decide it more than category positioning. Full mechanism-level comparison in [c1-research.md](#).

C1's July 2026 Launch Week pushed it well beyond this layer — Discovery & Context, Secrets & Credentials, and a narrow slice of Detection, in one coherent four-day rollout. Full breakdown and dedicated comparisons against Origin, 1Password, Artemis/CrowdStrike, and Tailscale Aperture in [c1-research.md](#).

The known gap: Okta, C1, and Lumos all answer "should this agent be here." None of them answer "does this agent actually understand what it's looking at" (that's Dosu) or "what did this agent actually do, in causal detail" (that's Origin) — though C1's Runtime Governance is the closest an Identity & Access player has come to touching that causal territory, via real-time enforcement rather than after-the-fact explanation.

Discovery & Context (Data vs. Knowledge)

Two companies solving the structurally identical problem for two different asset types:

	Cyera	Dosu
Discovers	Sensitive <i>data</i> across cloud, SaaS, on-prem	Project <i>knowledge</i> across GitHub repos, issues, docs
Answers	"What data do we have, and is it exposed?"	"What does this codebase mean, and is the agent's understanding correct?"
Why it matters now	AI agents touching data they shouldn't, or misclassifying sensitivity	AI agents touching codebases they don't understand, hallucinating fixes

Both are "you can't govern what you haven't discovered and correctly understood" — just applied to different objects.

Dosu's key nuance: the core grounding mechanism (indexing real project context to prevent hallucination) is separate from its A2A protocol work (letting other agents query that grounded knowledge). Grounding is the *how*; A2A is the *distribution*.

Detection

Behavioral, cross-domain detection and response. Artemis, CrowdStrike, and Splunk all belong here — full stop, regardless of deployment mechanism (an endpoint agent is just how the sensor gets deployed, not a separate capability).

Artemis vs. CrowdStrike, in one line: Artemis is an AI-native rebuild with a per-org dynamic detection model; CrowdStrike is an established platform (Falcon) with AI (Charlotte) layered on top, and its AI-activity visibility depends on vendors choosing to report through compliance channels rather than observing directly. Full comparison table in artemis-security-research.md.

Splunk's role: the dominant legacy SIEM every AI-native Detection player positions against — expensive at scale, complex, slow to adapt. The "AI-enabled bolt-on vs. AI-native rebuild" distinction that runs through this whole category starts with Splunk as the reference point.

The honest distinction with Origin: CrowdStrike/Artemis correlate what gets reported or observed. Origin observes what actually happens on the endpoint, independent of whether any vendor reports anything at all.

C1's Agentic Security & Intelligence is Detection-adjacent, not Detection proper — identity-configuration risk findings, not broad behavioral correlation. Closer to identity posture management than a SIEM. Full comparison in c1-research.md.

Telemetry Visualization & Aggregation

Not an identity-native layer — Grafana Labs isn't a security company, it's a general infrastructure observability platform (the LGTM stack: Loki for logs, Grafana for visualization, Tempo for traces, Mimir for metrics). It's placed here because it's a genuinely common destination layer: Detection, Endpoint, and even Identity & Access tooling often pipe their telemetry into a Grafana-style dashboard for human-readable correlation, rather than every layer building its own visualization from scratch.

Why it's worth tracking in this ecosystem specifically: Grafana's 2026 roadmap includes expanding its "All-in-One" suite into more advanced SOC/security analytics — a real signal the company is edging toward overlap with the Detection layer, not staying purely infrastructure-focused. Worth revisiting this placement as that expansion matures.

A live, relevant case study: Grafana itself suffered a real GitHub breach in May 2026 — a misconfigured GitHub Action (a "Pwn Request" vulnerability) exposed a token that let an attacker download the company's codebase, caught only via an internal canary token. The

compromise was later traced to the broader TanStack npm supply chain attack. It's a clean, current illustration of CI/CD pipeline risk — directly relevant to the same governance-gap thread running through this whole ecosystem map, just realized at a company that builds observability tooling rather than identity tooling.

Data Pipeline & Routing

Genuinely distinct from Telemetry Visualization, even though the two are adjacent in the observability stack. Grafana is a *destination* — where telemetry gets visualized and correlated after it arrives. [Cribl](#) is the *pipe* — it sits upstream of every destination, deciding what data gets routed where, in what shape, after being filtered and reduced. Cribl doesn't store or visualize data itself (aside from Cribl Lake as an optional cheap storage destination); it's infrastructure every other telemetry-consuming layer depends on.

The core value proposition: telemetry volume is growing ~29% annually per IDC, doubling roughly every 18 months while budgets stay flat. Cribl customers routinely cut data volumes 30-50% and can feed multiple destination tools (a SIEM, a cheap cold-storage bucket, a dashboard) from a single collection pass, without re-instrumenting anything.

The important industry signal: major EDR/Detection-layer vendors — CrowdStrike, Palo Alto Networks, and SentinelOne — all acquired competing pipeline tooling in Q4 2025 (Onum, Observe AI, and Chronosphere respectively) specifically to build direct ingestion lines into their own platforms. This creates a real structural tension: a vendor-owned pipeline is commercially incentivized to route the most data toward that vendor's own destination. A neutral pipeline like Cribl has no such incentive, since its business model depends on staying agnostic about where data ultimately lands — the same "neutral connective layer vs. platform-native alternative" tension that shows up elsewhere in this ecosystem (Tailscale's Network layer vs. platform-locked alternatives; C1's Agentic Vault vs. credential models tied to one identity platform).

Cribl on agentic AI specifically: CEO Clint Sharp has publicly framed agent-generated telemetry as needing the same routing and reduction discipline as any other observability data — just at much higher, less predictable volume as agent adoption scales.

Agent Construction

LangChain builds the orchestration and harness layer that turns a raw model into a working agent — prompts, tools, memory, planning, middleware. This is the "operating system" bet (per NVIDIA's Jensen Huang): most enterprises won't build their own agent orchestration from scratch, they'll build on top of a framework like LangChain.

Critical distinction: Agent Construction is explicitly NOT governance. LangChain and its competitors (CrewAI, AutoGen, LlamaIndex) have no native approval gates, policy enforcement, or audit trail before an agent acts on production systems. This is the seam where Identity & Access, Detection, and Hybrid Workforce Observability layers become necessary *alongside* a construction framework, not instead of it.

Endpoint Visibility

Tanium and Ivanti generate state and inventory truth — is the device compliant, what's installed, what changed, who touched it. Query-based, real-time at scale (Tanium's specific historic differentiator), feeding into patch/asset/vulnerability management.

[Axonius](#) sits here too, but with genuinely broader scope than pure endpoint — created the CAASM category, agentless discovery across devices, SaaS, cloud, identity, and OT/IoT, correlated into one system of record. Where Tanium/Ivanti are endpoint-first, Axonius is asset-first across the entire attack surface. Its Verified Assets push makes the same "garbage in, garbage out" argument already seen with Lumos's upstream data quality thesis. Full breakdown in axonius-research.md.

What this layer cannot do: explain *why* something happened, trace causal chains, or reason about intent. It can tell you a process ran or a file changed — it cannot tell you what an agent was trying to accomplish or what it read into context before acting.

Hybrid Workforce Observability

Origin's own language, used deliberately as the layer name: observability built for *"a workforce that is part-human and part-machine."*

This is a distinct layer from Endpoint Visibility, not a subset of it — the same **visibility vs. observability** split from monitoring/SRE practice. Visibility is state snapshots (what's installed,

what changed). Observability is causal tracing (why it happened, what the agent was trying to do). Full side-by-side comparison against Tanium/Ivanti in origin-research.md.

Why this layer is called out specifically: nobody else operates here with this depth. It sits adjacent to Detection and Identity & Access (an agent's commit identity not matching its SSO identity is a real identity governance signal, surfaced only because Origin sees the endpoint causally) but its core job — reconstructing intent and causality for both human and AI activity — is unique enough to deserve its own layer.

C1's Runtime Governance is the closest a governance-layer product has come to this territory — real-time enforcement per tool call, but requires the agent's traffic to route through C1's gateway. Origin requires no enrollment at all. Governance vs. observation, not a clean substitute either way — full comparison in c1-research.md.

Secrets & Credentials / AI Financial Visibility

1Password spans two distinct jobs, worth keeping separate: vaulting/rotation/NHI governance (Secrets & Credentials), and tracking AI token spend by vendor/team/user (AI Financial Visibility, launched July 2026). Full breakdown of both in 1password-research.md.

Critical distinction, in Origin's own words: *"Token bills tell you what you spent. [We] tell you what you bought."* Financial visibility is not behavioral or causal — a different layer solving a different pain (budget forecasting vs. causal governance).

C1's Agentic Vault vs. 1Password's Connect API: a genuine architecture difference — 1Password's Connect API still delivers the actual secret to the agent at runtime; C1's workload federation never gives the agent the raw secret at all. Two generations of the same problem. Full comparison in c1-research.md.

Cloud Posture

Wiz and Orca scan cloud environments for misconfigurations, exposed storage, overly permissive IAM policies. Out of scope for Network, Endpoint, or Observability layers entirely — a distinct surface.

IT Workflow Orchestration

ServiceNow doesn't generate ground-truth data — it consumes truth from other layers and turns it into automated workflows and remediation actions.

Two distinct ServiceNow-adjacent things, not to be confused:

	Now Assist AI Agents	ITOM AI Prime powered by Tanium
What it is	ServiceNow's own native AI agent product	A joint partnership — Tanium's real-time endpoint truth feeds ServiceNow's ITOM workflows for autonomous remediation
Ownership	Fully ServiceNow	Tanium (Endpoint Visibility) + ServiceNow (Orchestration), partnered

The partnership pattern here is the same shape as LangChain: **the orchestration layer needs ground truth from somewhere else to act on**. ServiceNow orchestrates; Tanium (or other endpoint tools) supplies the facts it orchestrates around.

The Governance Gap — The Thread Tying Every Layer Together

Across nearly every layer above, a consistent gap keeps appearing: **agent construction, orchestration, and even detection frequently lack native governance** — pre-action approval gates, policy enforcement, and audit trails before an agent acts.

- LangChain builds powerful agents but doesn't govern them
- Dosu makes agents smarter but doesn't govern them either
- Traditional Endpoint Visibility tools can't explain agent intent at all
- Even Detection tools depend heavily on what gets reported or observed after the fact

This gap is where Identity & Access (Okta, C1, Lumos), Hybrid Workforce Observability ([Origin](#)), and the emerging agentic governance thinking all converge. It's also the exact seam explored in the "Agentic vs. Human Identity" POV piece elsewhere in this KB — agents don't decide, they meet conditions, and the risk is unchecked execution of those conditions without enough visibility to catch when they're wrong.

Key Takeaways

- **ZTIA is a layer map, not a company catalog** — the same pain can be solved by different companies, and one company can operate in multiple layers
 - **Visibility and observability are not the same thing** — Endpoint Visibility (Tanium, Ivanti) tells you *what* happened; Hybrid Workforce Observability (Origin) tells you *why*
 - **Detection is Detection, regardless of deployment mechanism** — CrowdStrike and Artemis are the same layer; an endpoint agent is just how the sensor gets deployed
 - **Discovery & Context splits cleanly by asset type** — Cyera for data, Dosu for knowledge, same structural problem
 - **Agent Construction is explicitly not governance** — LangChain builds, it doesn't police what it builds
 - **Origin's differentiation is real and specific** — vendor-agnostic, causal, covers both human and AI activity, sees shadow agents nobody else can see
 - **The governance gap is the connective tissue** — nearly every layer eventually runs into "who approves this, who audits this, who can kill this" — and that gap is where identity security expertise becomes the differentiator
 - **C1 and Lumos are the closest head-to-head in Identity & Access** — both AI-native, both fast-deploying, execution and fit decide it more than category positioning
 - **Telemetry Visualization (Grafana) sits underneath, not beside, the security-native layers** — it's where other layers' data often gets seen, and it's a live example of CI/CD supply chain risk in its own right (the May 2026 GitHub breach)
 - **Data Pipeline & Routing (Cribl) is genuinely distinct from Telemetry Visualization** — the pipe vs. the destination, and the EDR vendors acquiring competing pipeline tools (CrowdStrike/Onum, Palo Alto/Observe AI, SentinelOne/Chronosphere) is a real structural signal about who wants to own the connective layer
 - **Axonius extends Endpoint Visibility into full asset attack-surface visibility (CAASM)** — broader than Tanium/Ivanti's endpoint-first scope, and its Verified Assets push makes the same upstream-data-quality argument already seen with Lumos, applied to asset inventory instead of identity governance
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Official References

Source	Link
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Cribl — What is an Observability Pipeline	https://cribl.io/blog/the-observability-pipeline/
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Source	Link
AI-Powered Remediation	
Tanium & ServiceNow — ITOM AI Prime Partnership	https://www.businesswire.com/news/home/20260506341909/en/Tanium-Combines-Forces-with-ServiceNow-to-Deliver-New-Autonomous-IT-Solution-Powered-by-Industry-Leading-Platforms
ConductorOne	https://www.conductorone.com
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Grafana Labs — GitHub Breach Coverage	https://www.securityweek.com/grafana-confirms-breach-after-hackers-claim-they-stole-data/